MILITARY SPECIFICATION

DEXTROMETHORPHAN HYDROBROMIDE AND GUAIFENESIN SYRUP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Dextromethorphan Hydrobromide and Guaifenesin Syrup (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.3)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

FSC 6505

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a pleasantly flavored, palatable syrupy liquid containing in each 5 ml, 100.0 mg of guaifenesin, 15.0 mg of dextromethorphan hydrobromide, and 1.4 percent alcohol, within the designated assay limits for the syrup.

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3.1.1 Additives. The finished preparation shall contain appropriate sweeteners and preservatives. Suitable coloring agent(s) or other ingredients may be added to assure compliance with the requirements stated herein. The coloring agent(s) and the preservatives in the product shall be in the amounts as approved by the Federal Food and Drug Administration (F.D.A.) which will make the preparation attractive and palatable, and will properly preserve the product.

3.1.2 Assay.

3.1.2.1 Guaifenesin. The finished preparation shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the required amount of guaifenesin (100.0 mg) in each 5 ml, when determined as specified in 4.5.1.

3.1.2.2 Dextromethorphan hydrobromide. The finished preparation shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the required amount of dextromethorphan hydrobromide (15.0 mg) in each 5 ml, when determined as specified in 4.5.2.

3.1.2.3 Alcohol. The finished preparation shall assay to contain not less than 1.1 percent and not more than 1.7 percent ethyl alcohol, by volume, when determined as specified in 4.5.3.

3.1.3 Identification. The retention time for the guaifenesin extracted from the syrup shall be the same as the retention time of Guaifenesin N.F. Standard when determined as specified in 4.5.4.

A copy of the standard graph shall be kept on file at the Defense Personnel Support Center for use if samples are submitted.

3.1.4 pH. The pH of the finished preparation shall be between 2.00 and 3.00 at 25° C., when determined potentiometrically, using the U.S.P. method.

3.1.5 Specific gravity. The specific gravity of the finished preparation shall be not less than 1.235 and not more than 1.255 at 25° C., when determined using a pycnometer, hydrometer, or specific gravity balance.

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3.1.6 Refractive index. The finished preparation shall have a refractive index of not less than 1.4300 and not more than 1.4400, when determined at 25° C., using an Abbe Refractometer or equivalent instrument giving comparable results.

3.1.7 Flavor and palatability. The finished syrup shall be palatable and pleasantly flavored with no unpleasant after-taste. Not later than the time specified for opening of bids or receipt of proposals, the offeror shall submit to the contracting officer six (6) individually packaged samples (each containing 4 fl oz) of Dextromethorphan Hydrobromide and Guaifenesin Syrup, representative of the product which the offeror proposes to furnish. Two (2) samples will be subjected to panel testing for a determination of palatability (see 4.5.5 - Palatability test). The remaining samples will be used by cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval as to palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification.

The requirement for submission of samples for use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the procuring activity on a previous procurement and the contracting officer determines that such product was previously procured and/or tested by the procuring activity and found to comply with the palatability requirements.

3.2 Active ingredients.

3.2.1 Guaifenesin and dextromethorphan hydrobromide. The guaifenesin and dextromethorphan hydrobromide used in the manufacture of the finished preparation shall be in accordance with the tests, standards, and requirements of the N.F., including any supplements or revisions, thereto.

3.2.2 Alcohol. The alcohol used in the manufacture of the finished preparation shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

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3.2.3 Other ingredients. All other ingredients used in the manufacture of the finished preparation shall be of U.S.P. or N.F. quality, or, if not included in either of these compendia, they shall be suitable for use in this cough syrup.

3.3 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the product, to the date of delivery to the Government.

3.4 Workmanship. Workmanship shall be first class throughout. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of syrup produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

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4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

Table I. Sampling

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled bottle
Minor	II	2.5	Filled bottle

4.3.1 For end item testing. Noncompliance with any of the following is cause for rejection.

Table II. End item testing

Characteristic	Requirement paragraph	Inspection level*	Minimum sample
Guaifenesin	3.1.2.1	S-2	5
Dextromethorphan hydrobromide	3.1.2.2	S-2	5
Alcohol	3.1.2.3	S-2	5
Identification	3.1.3	S-2	5
pH	3.1.4	S-2	5
Specific gravity	3.1.5	S-2	5
Refractive index	3.1.6	S-2	5
Flavor and palatability	3.1.7		

* Where possible, the sample shall be used for two or more tests

4.4 Examination. The finished syrup shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

Categories and defects*

Major
101 Syrup not free from foreign matter.
102 Syrup has crystal formation.
103 Syrup not homogeneous.

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Minor

201	Exterior of immediate container is not free from contents and foreign matter.
202	Label not free from defacing marks and tears.

* Inspection is not restricted to the classified possible defects listed above.

4.5 Tests.

4.5.1 Assay (guaifenesin in the finished preparation). Pipet 5 ml of the finished preparation into a separatory funnel. Add 5 ml of dilute sulfuric acid. Extract with four 25-ml portions of chloroform. Combine the extracts in a second separatory funnel. Wash the combined chloroform with 5 ml of purified water. Transfer the chloroform by filtering into a 100-ml volumetric flask. Dilute to volume with chloroform and mix well. Pipet 5 ml to another 100 ml volumetric flask. Dilute to volume with chloroform and mix. Using 1 cm silica cells, obtain the absorbance of the sample preparation at 275.5 nm. with chloroform as a blank. Balance the spectrophotometer to zero absorbance at 300 nm. with chloroform as a blank. Also obtain the absorbance of a solution of guaifenesin standard, N.F. similarly prepared, containing 0.04 mg of guaifenesin per ml.

Calculate percent of required amount of guaifenesin per 5 ml.

As an alternate to the above, the following assay methods shall be employed:

Method I

The method as described in the NF for the assay of Guaifenesin in Guaifenesin Syrup.

Method II

Standard.

Weigh 40.0 mg Guaifenesin, N.F. Standard into a 250-ml volumetric flask, and dilute to volume with chloroform. Pipet 20 ml into a 100-ml volumetric flask and dilute to volume with chloroform.

Procedure.

Pipet 2 ml ("to contain" pipet) of syrup (sample) into a 125-ml separator. Rinse the pipet with purified water. Add purified water to the separator until the volume is approximately 45 ml. Add 10 gram of anhydrous sodium sulfate and shake vigorously to dissolve the salt. Add 5 ml 10 percent sodium hydroxide and mix well. Wash the aqueous layer thoroughly with 50 ml isooctane (A.R.). Transfer the aqueous layer to a second

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125-ml separator. Wash the isooctane with 2 ml of purified water. Add this wash to the second separator. Discard the isooctane layer. Extract the aqueous layer with five 40-ml portions of chloroform. Filter the chloroform through Whatman No. 1 paper into a 250-ml volumetric flask. Rinse the paper and funnel with chloroform, dilute to volume, and mix. Pipet 20 ml of this solution into a 100-ml volumetric flask, q.s. with chloroform. Measure the absorbance of the photometer, or equivalent instrument giving comparable result, and chloroform as the blank. Employ baseline technique in measurement of the absorbance. Calculate percent of required amount of guaifenesin per 5 ml.

4.5.2 Assay (dextromethorphan hydrobromide in the finished preparation). Transfer 2 ml of the finished preparation into a separatory funnel, add 10 ml of 4 percent sodium hydroxide. Extract with four 25-ml portions of petroleum ether and combine in a second separatory funnel. Wash the ether with two 20-ml portions of purified water. Discard the water. Wash the ether with four 25-ml portions of 0.1N sulfuric acid. Discard the ether. Combine the acid washings by filtering into a 100 ml volumetric flask. Dilute to volume with 0.1N sulfuric acid, mix, and obtain its absorbance at 282 nm. Obtain the absorbance of a standard solution of dextromethorphan hydrobromide, N.F., similarly prepared containing 0.06 mg of dextromethorphan hydrobromide per ml.

Calculate percent of required amount of dextromethorphan hydrobromide per 5 ml.

4.5.3 Assay (alcohol in the finished preparation). Pipet into a volumetric flask exactly 50 ml of the finished preparation. Record the temperature of the sample and transfer it to a 500 ml Kjeldahl flask quantitatively using about 50 ml of purified water. Connect the flask to the Kjeldahl distillation apparatus, making sure all joints are tight. Distill about 48 ml of the liquid into a 50-ml volumetric flask immersed in an ice bath. Adjust the temperature of the distillate to the temperature of the original sample and fill to the mark with purified water. Mix well, then pour into a clean, tared pycnometer, and weigh. Calculate the percent of required amount of alcohol by referring to the alcohol table in the U.S.P.

As an alternate method to the above, an applicable, accurate reproducible, gas chromatographic method shall be employed.

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4.5.4 Identification (guaifenesin) in the finished preparation. Chromatographic determination.

Column:	6 foot - 10% Se-30 ON 80-100 S
Instrument:	F&M Model 700, Dual Flame Chromatograph or equivalent
*Instrument conditions for the F&M Model 700	Detector temperature - 270° C. Injection Port temperature - 280° C. Column temperature - 200° C. Operation - Isothermal. Carrier gas flow - 60 ml He per minute. Range - 10^2. Attenuation - 20 Sample Size - 3 ul.

*If another type chromatograph is employed, modify the instrument conditions and sample size accordingly.

Standard preparation. Dissolve exactly 500 mg of Guaifenesin N.F. Reference Standard in 25 ml of methylene chloride and mix.

Sample preparation. Pipet 25 ml of the sample preparation into a separatory funnel. Add 10 ml of dilute sulfuric acid. Extract with four 25-ml portions of methylene chloride. Combine the extracts. Evaporate to 10 ml. Transfer quantitatively to a 25-ml volumetric flask and dilute to volume with methylene chloride.

Procedure.

With the above described instrument and analysis conditions established, inject the sample into the chromatograph. After the sample peak is obtained and the instrument is clear, inject the prepared standard. Measure the retention time from the point of injection to the peak obtained for both sample and standard and compare.

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4.5.5 Palatability test. A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing (samples will be tested undiluted), coded, and served to panel members under controlled serving conditions, e.g., all samples will be of the same amount, and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied among panel members; an interval of at least five (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than the Palatability Standard,* when determined by the taste panel, using the following 9-point hedonic rating scale. The average rating of the sample shall be equal to, or greater than the average rating of the standard, similarly prepared and tested.

1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

* The Palatability Standard is available, upon a separate request to the contracting officer, Defense Personnel Support Center.

4.6 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

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Table III. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

NOTE: The above examination is only applicable to the unit, intermediate and exterior containers.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing four (4) fl. oz. as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twenty-four (24) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

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5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Ninety-six (96) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Ninety-six (96) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number.

Notes: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied the National Stock Number shall appear on the immediate container.

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5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.3 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-926-8985	DEXTROMETHORPHAN HYDROBROMIDE AND GUAIFENESIN SYRUP, 4 Fl. oz. (118 ml)

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1386

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-D-37231 DEXTROMETHORPHAN HYDROBROMIDE AND GUAIFENESIN SYRUP

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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